

CURRICULUM VITAE

SUMAN JYOTI DE

**Postdoctoral
Researcher**

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Suman Jyoti De

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Nationality: Indian

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Skills

Languages: Bengali (Mother Tongue), English, Hindi

Software: Mathematica, LaTeX, Overleaf

Programming Language: C++, C, Python

Other Skills: Good Analytical and Numerical Skills, Technical and Scientific Thinking, Effective Communication, and Writing Skills.

Education and Employments

McGill University/ Postdoctoral Researcher in Physics
Department, October 2023 - Present, Montreal, Canada

Harish-Chandra Research Institute/ Ph.D. in Theoretical Physics
Thesis: Study of Topological Phases in The Presence of Magnetic
Fields, August 2017 - September 2023, Prayagraj, India

**Harish-Chandra Research Institute, under Homi Bhabha
National Institute/** Master of Science (M. Sc.)
August 2015 - July 2017, Prayagraj, India

**Ramakrishna Mission Residential College Narendrapur affiliated
to University of Calcutta/** Bachelor of Science (B. Sc.)
August 2012 - July 2015, Kolkata, India

Awards and Recognitions

- **Infosys Award** for excellence in research (2021-2022).
- **Junior and Senior Research Fellow** of Harish-Chandra Research Institute, India (2017- 2023).
- Recipient of M. Sc. Scholarship (2015-2017) as **Integrated Ph. D. Scholar** from Homi Bhabha National Institute, India.
- Qualified for **Joint Entrance Screening Test (JEST)** and **Joint Admission Test (JAM)** for admission in M.Sc. in 2015, within top 100 ranks, India, 2015.
- Qualified for **National Graduate Physics Examination (NGPE)** as state topper, India, 2014.

- **Inspire Scholarship** (2012-2015) for excellence in Higher Education, Department of Science and Technology, India.

Publications and Ongoing works

List of Publications:

- **Flat-band formation and chiral superconductivity in driven topological insulators**, S J De, L Goutte, K Agarwal, T Pereg-Barnea, [arXiv: 2607.27335 \(2026\)](#).
- **Nanoscale defects as probes of time reversal symmetry breaking**, S J De, T Pereg-Barnea, K Agarwal, [PRX 16, 011001 \(2026\)](#).
- **Magnon transmission across $v=1|-1|1$ monolayer graphene junction as a probe of electronic structure**, S J De, S Rao, G Murthy, [PRB 110, 085417 \(2024\)](#).
- **Quantum spin Hall insulator in proximity with a superconductor: Transition to the Fulde- Ferrel- Larkin- Ovchinnikov (FFLO) state driven by a Zeeman field**, S J De, U Khanna, S Rao, S Das, [PRB 108, L161403 \(2023\)](#).
- **Global phase diagram of charge neutral graphene in the quantum Hall regime for generic interactions**, S J De, A Das, S Rao, R K Kaul, G Murthy, [PRB 107, 125422 \(2023\)](#).
- **Chiral detection of Majorana bound states at the edge of a quantum spin Hall insulator**, V Adak, A Mukhopadhyay, S J De, U Khanna, S Rao, S Das, [PRB 106, 045422 \(2022\)](#).
- **Emergence of spin-active channels at a quantum Hall interface**, A Saha*, S J De*, S Rao, Y Gefen, G Murthy, (* equal contributions) [PRB 103, L081401 \(2021\)](#).
- **Magnetic flux periodicity in second-order topological superconductors**, S J De, U Khanna, S Rao, [PRB 101, 125429 \(2020\)](#).

Ongoing works:

- **Loading Floquet bands using frequency chirps within fast quad pulse**, Ivan Martinez-Berumen, Suman Jyoti De, Tami Pereg-Barnea, and Bill Coish.
- **FFLO state and superconducting edge diode effects**, Sayak Bhowmik, Arijit Saha, and Suman Jyoti De.

Oral and Poster Presentations

- Oral presentation at the Canadian Association of Physics Congress meeting at Saskatoon, Canada, June 2025.
- Oral presentation in "Conference on Advances in

Topological Condensed Matter” at ICTP, Italy, Nov 2024.

- Poster presentation in INTERIQ meeting at Chateau Bromont, Canada, October 2024.
- Poster presentation in “Quantum Materials workshop” at the University of Toronto, Canada, May 2024
- Poster presentation at EPVWH conference at TIFR, Mumbai, India, January 2023.
- Oral presentation at QMAT conference at IIT Kanpur, U. P., India, September 2022.
- Oral presentation at APS satellite March Meeting (Hybrid) at ICTS, Bangalore, India, March 2022.
- Oral presentation at the APS March Meeting (Online), USA, March 2021.

Courses taken

Quantum Mechanics I & II, Classical Mechanics, Mathematica Methods I & II, Electrodynamics, Statistical Mechanics, Quantum Field Theory I & II, General Relativity, Condensed Matter Physics I & II, and Quantum Information during the M.Sc. program at Harish-Chandra Research Institute, India

Teaching Experiences

As a Teaching Assistant at Harish-Chandra Research Institute

- Electrodynamics course with Prof. Anirban Basu (2019).
- Advanced Statistical Mechanics course with Prof. Dileep Jatkar (2018).

Mentorship Experiences

As a co-supervisor with Prof. Tami Pereg-Barnea

- Benjamin Salewicz (2025-Present), M.Sc. student of McGill University, Canada, on the project titled **Population control of Floquet bands using system-bath couplings**.

As a co-supervisor with Prof. Sumathi Rao

- Ananya Janardhanan (2021-2022), M.Sc. student of IISER Bhopal, India, on the project titled **Edge reconstruction in quantum Hall systems using Monte Carlo methods**.

Other Academic Experiences

- Actively working as a reviewer in the Physical Review Journals and Nature Physics.